BAMEE301 - COMPUTATIONAL ENGINEERING

MODULES 3 AND 4 - ASSIGNMENT

Instructions: Attempt the problems by showing the governing numerical formulation, intermediate computational steps, convergence/error checks wherever applicable, and a brief engineering interpretation of the result. Unless otherwise stated, retain at least four significant digits in intermediate calculations.

MODULE 3 - APPROXIMATION, DIFFERENTIATION, AND INTEGRATION

1. Newton Divided-Difference Interpolation

1. The dynamic viscosity of a high-temperature gearbox lubricant is measured at unequally spaced temperatures: T (°C) = [25, 40, 60, 85, 115] and μ (mPa·s) = [162.0, 104.8, 61.5, 31.8, 16.4]. Construct the complete Newton divided-difference table, obtain the interpolation polynomial using all five data points, estimate μ at 68 °C, and comment on whether a single linear interpolation is adequate over the complete range.

2. Calibration of an exhaust-manifold thermocouple gives E (mV) = [2.10, 4.35, 6.20, 8.75, 11.40] at T (°C) = [95, 188, 272, 401, 548]. Develop the Newton divided-difference polynomial and estimate the temperature corresponding to E = 7.20 mV. Evaluate the polynomial at one listed calibration point as a consistency check.

2. Lagrange Interpolation

3. The coefficient of friction of a brake-pad material is μ = [0.42, 0.47, 0.51, 0.46, 0.37] at T (°C) = [80, 140, 200, 280, 360]. Using all five points, estimate μ at 225 °C by Lagrange interpolation. Repeat using only the three nearest points and quantify the percentage difference.

4. An electric-vehicle battery module gives temperature rises ΔT (°C) = [2.1, 4.8, 8.9, 17.6, 30.5] at discharge currents I (A) = [1.0, 2.0, 3.0, 4.5, 6.0]. Estimate ΔT at 3.6 A using a quadratic Lagrange polynomial based on the three nearest points and a fourth-order polynomial using all five points. Compare the estimates and comment on sensitivity to polynomial order.

3. Cubic Spline Interpolation

5. The measured force-displacement response of an elastomeric vibration mount is x (mm) = [0, 5, 10, 15] and F (kN) = [0.0, 2.8, 7.5, 15.2]. Construct a natural cubic spline, obtain the cubic equation for each interval, estimate F at x = 7.5 mm, verify first-derivative continuity at the internal knots, and interpret the local slope as tangent stiffness.

6. Cam-follower measurements are θ (deg) = [0, 15, 30, 45] and displacement s (mm) = [0.0, 6.2, 14.8, 20.0]. Construct a natural cubic spline, estimate displacement and slope at θ = 22.5°, identify the segment used, and explain why a piecewise cubic representation is preferable to one high-order polynomial.

4. Linear Regression and Curve Fitting

7. A tensile specimen gives strain ε (×10^-3) = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0] and stress σ (MPa) = [102, 201, 304, 397, 506, 598]. Using least-squares linear regression, determine $\sigma = a + b\varepsilon$, estimate Young's modulus from the slope, calculate R^2, and comment on the physical significance of the intercept.

8. Coast-down testing gives ω (rad/s) = [60, 90, 120, 150, 180, 210] and resisting torque T (N·m) = [8.4, 10.1, 12.5, 14.0, 16.2, 17.8]. Fit $T = a + b\omega$, estimate T at 145 rad/s, determine R^2, and interpret a and b in terms of Coulomb-type and viscous resistance.

5. Polynomial Regression

9. A progressive suspension spring gives x (mm) = [0, 10, 20, 30, 40, 50] and F (kN) = [0.2, 2.0, 5.1, 9.7, 16.1, 24.6]. Fit $F = a0 + a1x + a2x^2$ by least squares, estimate F at x = 32 mm, calculate R^2, and compare tangent stiffness at 32 mm with average stiffness over 20-40 mm.

10. Wind-tunnel data are V (m/s) = [10, 15, 20, 25, 30, 35] and drag F (N) = [42, 91, 158, 248, 357, 489]. Fit both a linear and a quadratic model, compare residual sum of squares and R^2, estimate drag at 27 m/s, and justify which model is mechanically more appropriate.

6. Numerical Differentiation - Forward, Backward, and Central Differences

11. A beam gives x (m) = [0, 0.25, 0.50, 0.75, 1.00] and deflection y (mm) = [0.000, 0.920, 2.180, 3.960, 6.420]. Estimate slope at x = 0 using a second-order forward formula, at x = 1.0 m using a second-order backward formula, and at x = 0.5 m using a central formula. Determine curvature at x = 0.5 m and, for E = 200 GPa and I = 6.0×10^-6 m^4, estimate bending moment.

12. Machine-foundation displacement is sampled at Δt = 0.01 s: t (s) = [0.00, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06], x (mm) = [0.000, 0.412, 0.775, 1.041, 1.176, 1.158, 0.982]. Use forward, backward, and central differences to estimate velocity at t = 0.03 s and central difference to estimate acceleration. Compare the velocity estimates and discuss noise amplification.

7. Numerical Integration - Trapezoidal Rule

13. A crash-energy absorber gives x (mm) = [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50] and F (kN) = [0.0, 1.1, 2.6, 4.8, 7.9, 12.0, 17.2, 23.6, 31.5, 41.0, 52.3]. Estimate absorbed energy using the composite trapezoidal rule. Repeat using every alternate data point and determine the percentage change. Discuss the influence of step size.

14. A variable aerodynamic drag force is F(x) = 900e^(0.35x) + 120sin(2x) N over $0 \le x \le 2$ m. Use the composite trapezoidal rule with n = 4 and n = 8 subintervals to estimate work. Compare the estimates and use their difference as a discretization-sensitivity measure.

8. Numerical Integration - Simpson's Rule

15. Using the crash-energy absorber data in Problem 13, estimate absorbed energy using Simpson's 1/3 rule. Compare with the trapezoidal estimate using the same spacing, calculate the percentage difference, and explain why the two methods need not give identical results for nonlinear experimental data.

16. The resisting torque of a rotary damper is $T(\theta) = 45 + 18\sin\theta + 12\sin(3\theta) + 0.8\theta^2$ N·m over $0 \le \theta \le \pi$. Estimate work dissipated using Simpson's 1/3 rule with n = 6 and n = 12 subintervals. Compare the estimates and comment on convergence.

9. Gaussian Quadrature

17. The axial force in a nonlinear actuator is F(x) = 1800e^(-0.6x)sin(2.5x) + 700 N over $0 \le x \le 1.2$ m. Estimate work using 1-point, 2-point, and 3-point Gauss-Legendre quadrature. Compare successive estimates and explain why finite Gauss quadrature is not exactly exact for this non-polynomial integrand.

18. For a tapered finite element, $I = \int$ from 0 to 2 of $(3 + 2x + x^2 + 0.5x^4)\,dx$. Evaluate I using 1-point, 2-point, and 3-point Gauss-Legendre quadrature after mapping to [-1,1]. Compare with the analytical integral and relate the result to the 2n-1 exactness property.

MODULE 4 - PDE CLASSIFICATION AND INITIAL-VALUE PROBLEMS

10. Classification of PDEs - Parabolic, Elliptic, and Hyperbolic

19. Classify using the discriminant criterion and identify one mechanical-engineering application for each: (i) $T_{xx} + 4T_{xy} + T_{yy} = 0$, (ii) $4u_{xx} + 4u_{xy} + u_{yy} = 0$, and (iii) $u_{xx} - 4u_{yy} = 0$. Explain what each classification implies about the physical character of the problem.

20. Consider (i) steady 2-D heat conduction $\nabla^2 T = 0$, (ii) transient 1-D heat conduction $T_t = \alpha T_{xx}$, and (iii) longitudinal wave propagation $u_{tt} = c^2 u_{xx}$. Rewrite each in a suitable second-order PDE form, classify it, state the initial/boundary data normally required, and distinguish equilibrium, diffusion, and wave propagation.

11. Initial-Value Problems - Euler's Method

21. A flywheel with $J = 3\ \text{kg·m}^2$ is driven by $T(t) = 24 + 8\sin(2t)$ N·m and experiences viscous resistance 5ω. Starting from $\omega(0) = 1$ rad/s, formulate the first-order IVP and use explicit Euler with $h = 0.1$ s to estimate ω up to $t = 0.5$ s. Repeat with $h = 0.05$ s and compare final values.

22. A lumped thermal model of a machine casing is $dT/dt = -0.08(T - 25) + 3e^{(-0.5t)}$, $T(0) = 90$ °C. Use Euler's method with $h = 0.5$ min to estimate T up to 3 min. Repeat with $h = 0.25$ min, compare $T(3)$, and explain the physical meaning of each ODE term.

12. Runge-Kutta Methods

23. A damped actuator response is $dy/dt = -1.5y + 2\sin(3t)$, $y(0) = 0.8$. Use classical RK4 with $h = 0.1$ s to estimate y at $t = 0.4$ s. Tabulate k1, k2, k3, and k4 for every step and compare with an Euler solution using the same h.

24. Rotor speed satisfies $d\omega/dt = 12 - 0.8\omega - 0.04\omega^2$, $\omega(0) = 2$ rad/s. Use RK2 (midpoint) and RK4 with $h = 0.2$ s to estimate ω at $t = 1.0$ s. Compare predictions and discuss the influence of method order for this nonlinear IVP.

13. Adams-Bashforth Predictor Method

25. The response satisfies $dy/dt = -2y + \cos t$. Starting values are $y(0.0) = 1.000000$, $y(0.1) = 0.900000$, $y(0.2) = 0.819500$, and $y(0.3) = 0.755019$. Using the four-step Adams-Bashforth predictor, estimate $y(0.4)$ and $y(0.5)$. Clearly evaluate derivative history and explain why a multistep method needs starting values.

26. A rotor-speed model is $d\omega/dt = 8 + 2\sin t - 0.6\omega$, $h = 0.1$ s. Given $\omega(0.0) = 2.0000$, $\omega(0.1) = 2.6650$, $\omega(0.2) = 3.2921$, and $\omega(0.3) = 3.8865$, use the four-step Adams-Bashforth method to predict $\omega(0.4)$ and $\omega(0.5)$. Comment on how starting-value error can influence later predictions.

14. Adams Predictor-Corrector Method

27. For $dy/dt = -y + 0.5\sin(2t)$, $h = 0.1$ s, starting values are $y(0.0) = 1.000000$, $y(0.1) = 0.909675$, $y(0.2) = 0.836857$, and $y(0.3) = 0.779050$. Use the four-step Adams-Bashforth formula to predict $y(0.4)$, then apply the Adams-Moulton corrector. Perform a second correction and quantify the change between successive corrected values.

28. A thermal response satisfies $dT/dt = -0.12(T - 30) + 4\cos(0.5t)$, $h = 0.5$ min. Given $T(0) = 85.000$, $T(0.5) = 83.120$, $T(1.0) = 81.392$, and $T(1.5) = 79.818$ °C, use an Adams-Bashforth predictor and Adams-Moulton corrector to estimate T(2.0 min). Perform correction until successive corrected values differ by less than 0.001 °C and report the number of corrections required.